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PineCone Vector Database
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First, what problem does Pinecone solve?

Imagine you have a company chatbot.

You have thousands of documents:

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HR policies
Product manuals
FAQs
Employee documents
Customer support articles
Technical documentation
```

A user asks: "How many days of casual leave can I take?"

A normal database might search for the exact words "casual leave".

But GenAI applications need something smarter.

The user might ask: "How many days can I take off for personal reasons?"

Even though the words are different, the meaning is similar.

This is where vector databases such as Pinecone become useful.

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2. What is Pinecone?
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** Pinecone is a vector database designed for AI applications.

** Its main job is to store embeddings and quickly find information that is semantically similar to a user's question.

** Pinecone stores numerical representations of information and helps you find similar information very quickly.

Ex :

```
"I love programming"
  â†“
Embedding Model
  â†“
[0.21, -0.45, 0.78, 0.12, ...]
```

** This list of numbers is called an embedding vector.

** The vector represents the meaning of the text.

Sentence-1 : "I love programming"

Sentence-2 : "Software development is my passion"

may produce vectors that are relatively close to each other because their meanings are similar.

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Real-Time Example: Customer Support Chatbot
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Imagine an e-commerce company has these documents:

```
return_policy.txt
refund_policy.txt
shipping_policy.txt
payment_policy.txt
product_manual.txt
```

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Important Pinecone Terminology
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- 1) Index
- 2) Vector
- 3) ID
- 4) Meta Data
- 5) Namespaces

** An index is where your vectors are stored and searched.

** Vector means numerical representation of our data

** Every vector needs a unique ID.

** Metadata means Extra information associated with the vector. Metadata is extremely useful for filtering.

** A namespace allows you to logically separate data inside an index.

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RAG with PineCone Vector DB Project setup
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Step-1 :: Create project folder

Step-2 :: Go inside project folder, Create virtual environment and activate it

```
cd <project-folder-name>
python -m venv venv
venv\Scripts\activate
```

Step-3 :: Create requirements.txt file inside the project with below libraries and install them using 'pip'

```
...
openai
pinecone
python-dotenv
streamlit
```

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'''

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$ pip install -r requirements.txt

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## Step-4: Create .env file and configure open AI API key & PineCone API Key

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    OPENAI_API_KEY=<key-goes-here>
    PINECONE_API_KEY = <key-goes-here>

```

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## Step-5: Create "documents\company.txt" file with required data to load into vector db

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## Step-6: Create "ingest.py" file to read file data and store into vector db (Run it)

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    $ python ingest.py

```

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## Step-7: Create "rag.py" file to perform "Retrieve + Augment + Generation"

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    $ python rag.py

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## Step-8: Create "app.py" file with Streamlit UI

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    $ streamlit run app.py

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=====Understanding Output=====

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Results: QueryResponse(matches=[ScoredVector(id='chunk-4', score=0.21221517, values=[],
metadata={'text': 'The company provides health insurance to all full-time employees.'}),
ScoredVector(id='chunk-3', score=0.209931836, values=[], metadata={'text': 'Employees receive 20
paid leaves every year.'}), ScoredVector(id='chunk-5', score=0.136570811, values=[],
metadata={'text': 'The notice period for employees is 30 days.'})], namespace='',
usage=Usage(read_units=1, write_units=None), response_info=ResponseInfo(raw_headers={'date': 'Fri,
11 Sep 2026 02:39:06 GMT', 'content-type': 'application/json', 'content-length': '432',
'connection': 'keep-alive', 'x-pinecone-max-indexed-lsn': '7', 'x-pinecone-request-latency-ms':
'277', 'x-envoy-upstream-service-time': '50', 'x-pinecone-response-duration-ms': '279', 'grpc-
status': '0', 'server': 'envoy'})))

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matches=[ScoredVector(id='chunk-4', score=0.21221517, values=[], metadata={'text': 'The company
provides health insurance to all full-time employees.'}), ScoredVector(id='chunk-3',
score=0.209931836, values=[], metadata={'text': 'Employees receive 20 paid leaves every year.'}),
ScoredVector(id='chunk-5', score=0.136570811, values=[], metadata={'text': 'The notice period for
employees is 30 days.'})]

```

```

-----
[The company provides health insurance to all full-time employees,
Employees receive 20 paid leaves every year,
The notice period for employees is 30 days]

```

```

-----
The company provides health insurance to all full-time employees

```

```

Employees receive 20 paid leaves every year

```

```

The notice period for employees is 30 days

```

```

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Assignment :

Develop a RAG Based application with Pinecone Vector DB to process PDF Docs data.

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